



ViAmpleHate

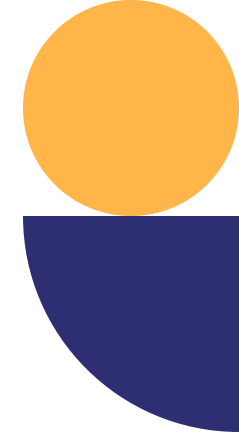
A Proposed AmpleHate- and PhoBERT-based Approach for Vietnamese Hate

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Abstract

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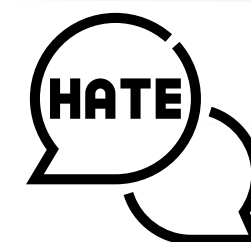
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ViAmpleHate

A Proposed AmpleHate- and PhoBERT-based Approach for
Vietnamese Hate Speech Detection





Abstract

Detecting hate speech on Vietnamese social media is difficult because hateful intent often appears implicitly through informal group references, slang, sarcasm, and indirect insinuation rather than explicit named entities. Although AmpleHate shows that modeling the relation between an utterance and its target can improve implicit hate detection, its English NER and English-oriented target categories transfer poorly to Vietnamese: on our training data, English NER finds usable targets in only about 0.09% of comments, causing the model to behave almost like a plain sentence classifier.

We propose ViAmpleHate, a PhoBERT-based Vietnamese adaptation of AmpleHate that replaces English NER with Vietnamese NER and curated target-cue and attack-cue banks. This raises target coverage from $\approx 0.09\%$ to $\approx 19\text{--}45\%$, depending on the dataset, and models explicit target, implicit context, and attack relations through relation-bank attention with an instance-adaptive gate. On ViHSD and VOZ-HSD, ViAmpleHate improves macro-F1 and HATE-class F1 over AmpleHate, TF-IDF, BiLSTM, and PhoBERT-CNN baselines, with the clearest gain on VOZ-HSD, where HATE-F1 increases by +0.0508; results on ViHSD are smaller and reported as preliminary.



1. Introduction

Background & Challenges

Vietnamese hate speech detection is challenging because social-media language is noisy, informal, and often intentionally distorted.



Hate is rarely expressed through explicit named entities; instead, it targets groups indirectly through pronouns, kinship terms, regional labels, slang, sarcasm, or informal address forms. In addition, the datasets are highly imbalanced, with the HATE class making up only about one tenth of comments.



#1. Introduction

Proposed ViAmpleHate

We propose ViAmpleHate, a PhoBERT-based adaptation of AmpleHate for Vietnamese hate speech detection.

Main Idea

It replaces English NER with Vietnamese NER and curated target-cue banks, increasing target coverage from about 0.09% to 18.8–20.0%.

Other enhancements

ViAmpleHate also separates target and attack cues, uses relation-bank attention with adaptive fusion, and combines weighted cross-entropy with contrastive loss to improve HATE/NON-HATE classification.

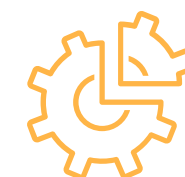


2. Related Work



Hate speech detection and implicit hate

Automatic hate speech detection has progressed from lexicon-based methods to deep learning and pretrained transformers. However, implicit hate remains difficult because hostility is often expressed through sarcasm, stereotypes, coded references, or shared social knowledge rather than explicit slurs.



Contrastive Learning for Text Classification

Contrastive learning helps bring same-class representations closer and push different-class representations apart. ViAmpleHate combines weighted cross-entropy with a contrastive auxiliary objective to improve separation between the frequent NON-HATE class and the rare HATE class.



Vietnamese Hate Speech Detection

Vietnamese hate speech often depends on informal target references such as pronouns, group nouns, and address forms. Existing datasets show the importance of target information, but ViAmpleHate does not require span-level annotation; it uses Vietnamese NER and cue banks to locate target and attack positions.



2. Related Work



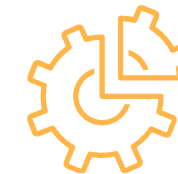
AmpleHate

AmpleHate detects implicit hate by modeling the relationship between a sentence and its candidate targets. It uses NER to extract target tokens and applies target-aware attention before classification. However, it relies on English NER, fixed relation injection, and focuses only on target cues while ignoring attack cues.



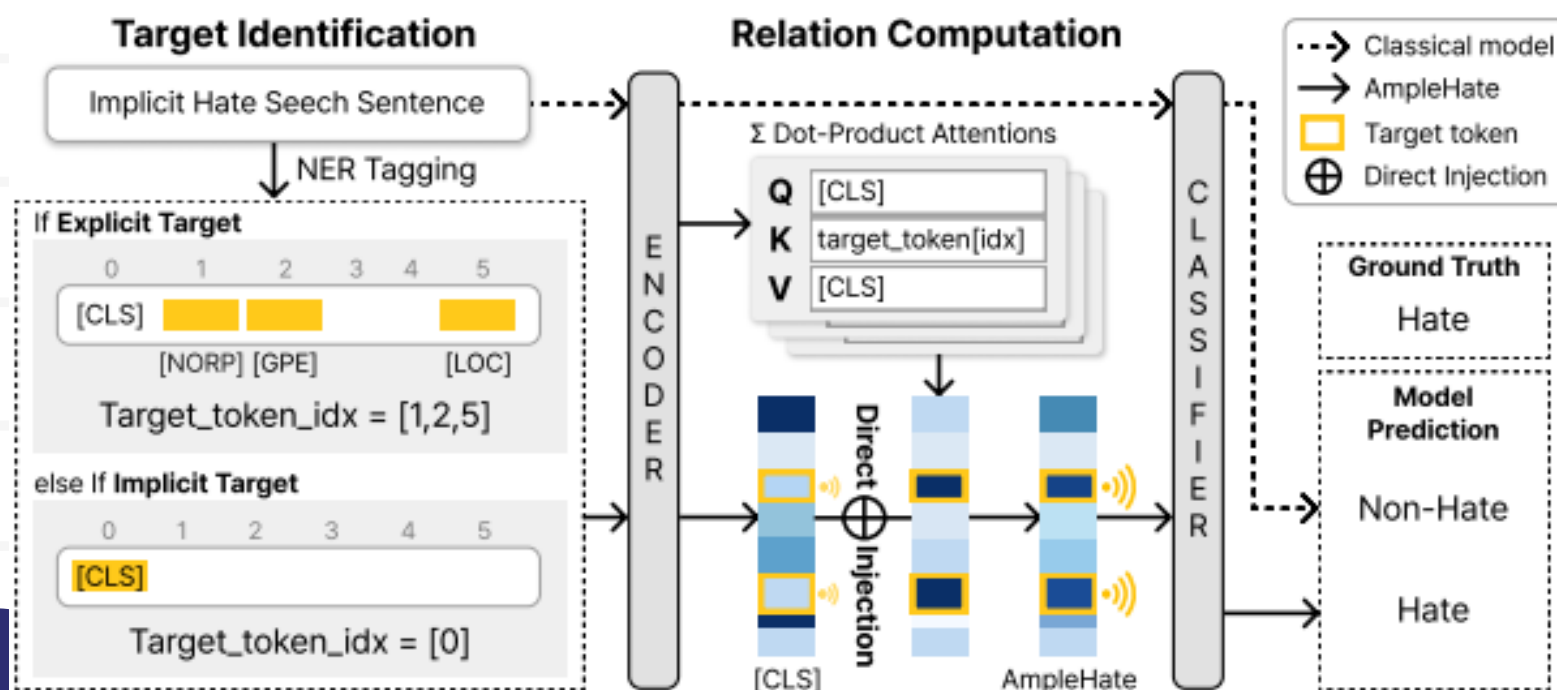
Fine-tuned Models for
vinai/phobert-base

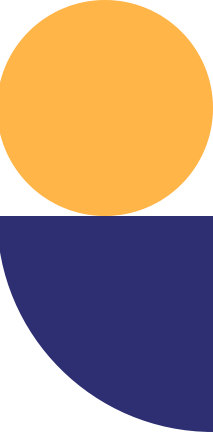
huggingface.co/models



PhoBERT and Vietnamese Pretrained Models

PhoBERT is a strong pretrained encoder for Vietnamese text classification. Since it is trained on word-segmented Vietnamese text, word segmentation is applied before tokenization. ViAmpleHate uses PhoBERT-base as the shared encoder so improvements come from target-aware modeling rather than the backbone.





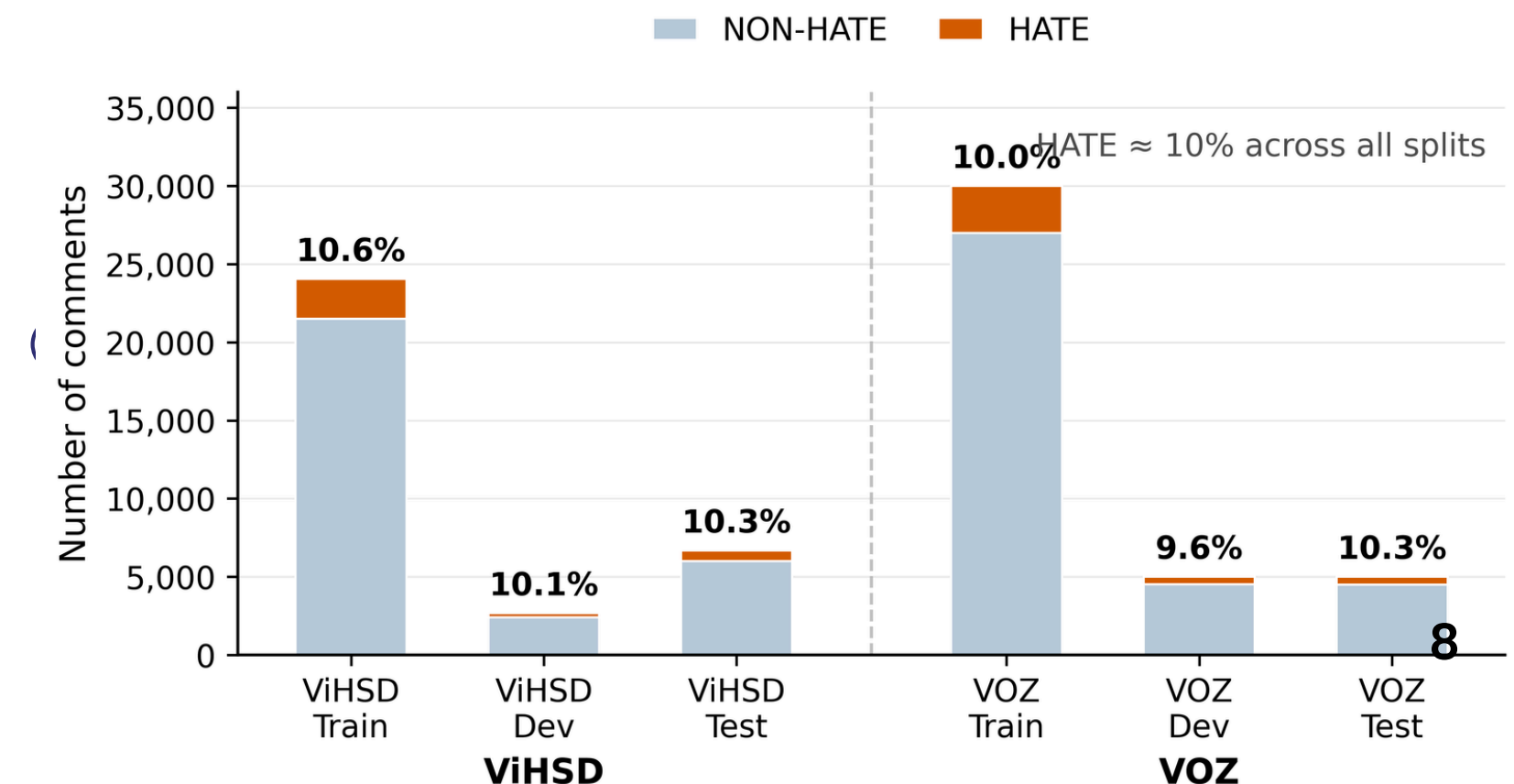
ViHSD VOZ-HSD

Dataset	Split	NON-HATE	HATE	Total
ViHSD	Train	21,492	2,556	24,048
ViHSD	Dev	2,402	270	2,672
ViHSD	Test	5,992	688	6,680
VOZ	Train	26,993	3,007	30,000
VOZ	Dev	4,520	480	5,000
VOZ	Test	4,487	513	5,000

The task is converted into binary classification by merging CLEAN and OFFENSIVE into NONHATE, while HATE remains the positive class. No examples are removed or subsampled.

3. Dataset

Comments are normalized through lowercasing, URL/symbol removal, elongated-word reduction, teencode normalization, and emoji mapping. They are then word-segmented and tokenized using PhoBERT.





3. Dataset

```
TARGET_CUES = [  
    'bạn', 'thằng', 'con', 'đứa', 'tụi', 'đám', 'lũ',  
    'mấy', 'loại', 'người', 'dân', 'bên',  
    'hắn', 'chúng', 'họ', 'nó',  
]  
  
ATTACK_CUES = [  
    'ngu', 'đần', 'ngu-ngốc', 'khùng', 'điên', 'hèn', 'nhục',  
    'ăn_bám', 'kỳ_sinh', 'phản_quốc', 'vô_học', 'man_rợ',  
    'cút', 'xéo', 'câm_miệng',  
    'giết', 'chém', 'đánh',  
    'ghét', 'khinh', 'chửi',  
    'vô_văn_hóa', 'thấp_hèn', 'đáng_chết',  
]
```

All cues are normalized, word-segmented, tokenized, and matched as full token sequences. The final banks contain 16 target cues and 24 attack cues, but may be limited by training-data bias and low coverage of new slang.

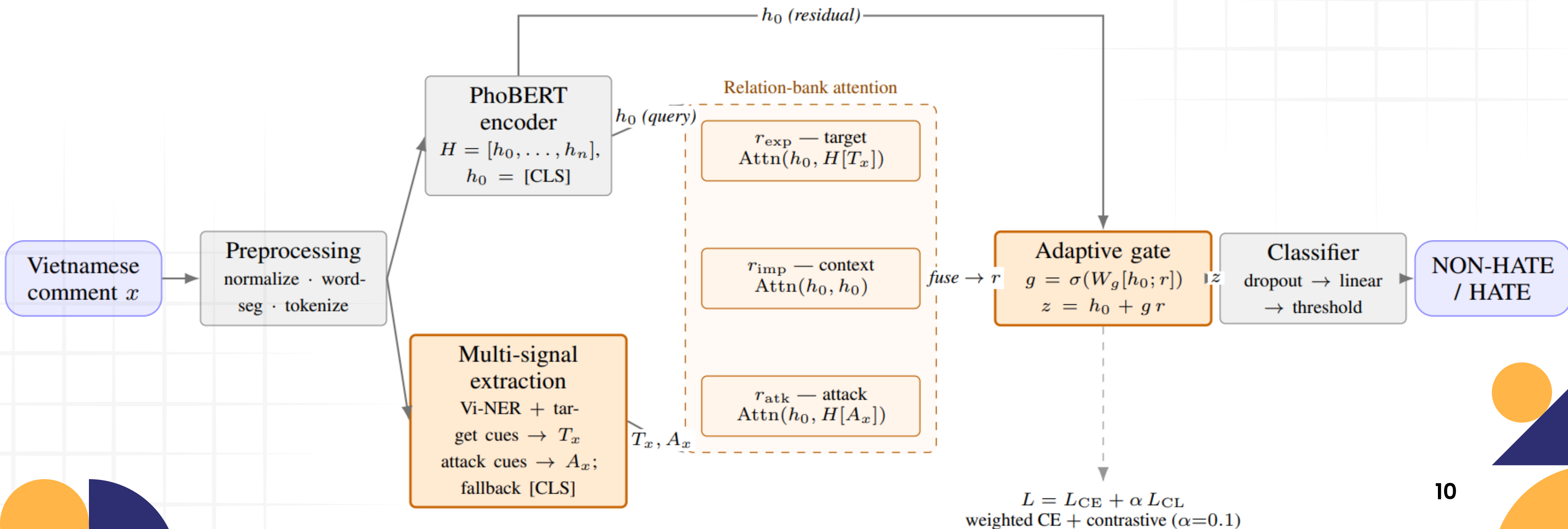


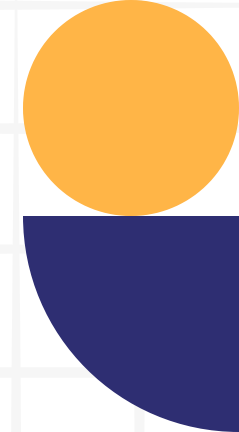
ViAmpleHate uses two manually curated cue banks: target cues and attack cues. Target cues refer to people or groups, while attack cues include insults, threats, dehumanizing terms, and negative evaluations.





4. Methodology





5. Experiments

Baseline models

Machine Learning

- TFIDF + LR and TF-IDF + SVM

Neural Network Model with Static Embeddings

- BiLSTM + fastText

<https://fasttext.cc/docs/en/crawl-vectors.html>

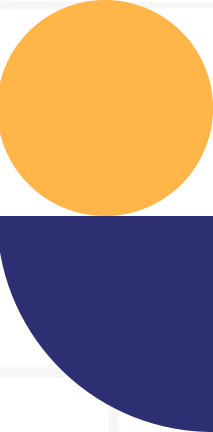
PhoBERT-based Deep Learning Models

- PhoBERT-CNN

Original Approach from related works

- AmpleHate-PhoBERT,

['dbmdz/bert-large-cased-finetuned-conll03-english'](#)



5. Experiments

Proposed model

VAmplateHate – PhoBERT

encoder

vinai/phobert-base

huggingface.co/models

Vietnamese NER

 NlpHUST/ner-vietnamese-electra-base



Token Classification



Transformers



PyTorch



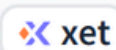
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Model card



Files and versions



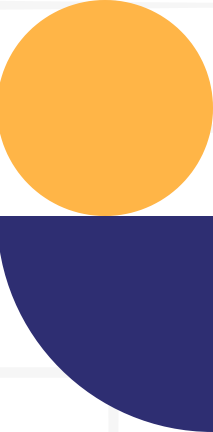
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Vietnamese NER

Parameter	Value
Encoder	vinai/phobert-base (768-d)
Vietnamese NER	NlpHUST/ner-vietnamese-electra-base
max_len	256
LR (encoder / head)	2e-5 / 5e-5
Dropout	0.1
Effective batch	32 (16 × grad-accum 2)
Epochs	up to 8 (best by val macro-F1)
α (contrastive)	0.1
Class weights w_c	inverse freq. $N/(C n_c)$
Contrastive margin m	0.5
Label smoothing	0.05
Seed / runs	42 / single run



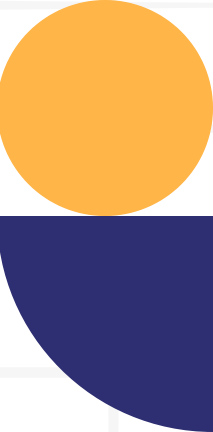
5. Experiments

Model	Acc.	Mac-F1	HATE-F1
TF-IDF + LR	0.8910	0.7393	0.5404
TF-IDF + SVM	0.9126	0.7131	0.4739
BiLSTM + fastText	0.8454	0.7072	0.5060
PhoBERT-CNN	0.8945	0.7571	0.5745
AmpleHate (baseline)	0.9175	0.7792	0.6045
ViAmpleHate	0.9205	0.7819	0.6081
$\Delta base$	$+.0030$	$+.0027$	$+.0036$

Table 3: Results on ViHSD (test). Best per column in bold (accuracy shown for reference). Single run (seed 42).

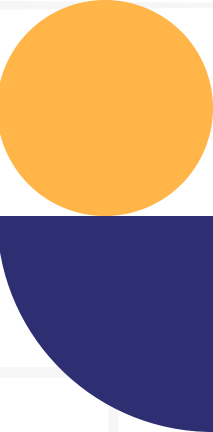
Model	Acc.	Mac-F1	HATE-F1
TF-IDF + LR	0.9453	0.7745	0.5783
TF-IDF + SVM	0.9641	0.7831	0.5850
BiLSTM + fastText	0.8650	0.6712	0.4187
PhoBERT-CNN	0.9623	0.8150	0.6500
AmpleHate (baseline)	0.9643	0.8185	0.6557
ViAmpleHate	0.9420	0.8371	0.7065
$\Delta base$	$-.0223$	$+.0186$	$+.0508$

Table 4: Results on VOZ-HSD (test). Best per column in bold among models (accuracy shown for reference; the baseline has the highest accuracy). Single run (seed 42).



5. Experiments

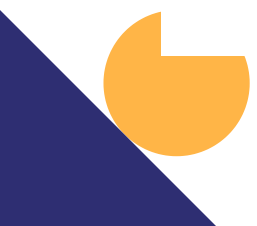
Dataset	Model	Accuracy	Macro Precision	Macro Recall	Macro F1	HATE Precision	HATE Recall	HATE F1
ViHSD	TF-IDF + SVM (Tuned)	0.9126	0.78	0.68	0.7131	0.62	0.38	0.4739
ViHSD	TF-IDF + LR (Tuned)	0.891	0.72	0.77	0.7393	0.48	0.62	0.5404
ViHSD	BiLSTM - FastTextVi	0.8454	0.6735	0.8115	0.7072	0.377	0.7689	0.506
ViHSD	PhoBert CNN	0.8945	0.7271	0.8048	0.7571	0.4912	0.6919	0.5745
ViHSD	PhoBert AmpleHate Origin	0.9175	0.7762	0.7823	0.7792	0.5972	0.6119	0.6045
ViHSD	PhoBert ViAmpleHate	0.9205	0.7859	0.7781	0.7819	0.6177	0.5988	0.6081
VOZ-HSD	TF-IDF + SVM (Tuned)	0.9641	0.85	0.74	0.7831	0.72	0.49	0.585
VOZ-HSD	TF-IDF + LR (Tuned)	0.9453	0.73	0.84	0.7745	0.48	0.73	0.5783
VOZ-HSD	BiLSTM - FastTextVi	0.865	0.633	0.8851	0.6712	0.2721	0.9076	0.4187
VOZ-HSD	PhoBert CNN	0.9623	0.8021	0.8292	0.815	0.6217	0.6809	0.65
VOZ-HSD	PhoBert AmpleHate Origin	0.9643	0.8159	0.8211	0.8185	0.6501	0.6615	0.6557
VOZ-HSD	PhoBert ViAmpleHate	0.942	0.8492	0.8261	0.8371	0.7347	0.6803	0.7065



5. Experiments

Component	Baseline AmpleHate-PhoBERT	ViAmpleHate-PhoBERT (proposed)
Encoder	PhoBERT-base	PhoBERT-base
Target extraction	English NER	Vietnamese NER + target-cue bank
Target coverage	$\approx 0.09\%$ of train (21/24,048)	$\approx 18.8\text{--}20.0\%$ via Vietnamese cues
Attack signal	Not modeled	Separate attack-cue bank
Attention	One HeadAttention	Three channels: target / implicit / attack
Attention computation	Batch-level matmul (mixes samples)	Per-sample bmm + masking
Fusion	Fixed injection $h_0 + e r$ ($e=1.0$)	Relation bank + adaptive gate $h_0 + g r$
Loss	Weighted CE	Weighted CE + contrastive ($\alpha=0.1$)
Max length	128	256
Batch	16	$16 \times \text{grad-accum } 2$ (eff. 32)
NER at evaluation	Off ($\Rightarrow$ [CLS] fallback)	On, consistent across splits

Table 2: Architecture and configuration comparison between the baseline and the proposed model.





01 Main result

- ViAmpleHate achieves the best Macro-F1 and HATE-F1 on both ViHSD and VOZ-HSD.
- On ViHSD, the improvement over AmpleHate is small: Macro-F1 increases by 0.0027 and HATE-F1 by 0.0036.
- On VOZ-HSD, the gain is much clearer: Macro-F1 increases by 0.0186 and HATE-F1 by 0.0508.
- The VOZ-HSD accuracy decreases, but minority-class performance improves, showing that accuracy is less informative under strong class imbalance.

Data	Class	P	R	F1
ViHSD	NON-HATE	0.9541	0.9574	0.9558
ViHSD	HATE	0.6177	0.5988	0.6081
VOZ	NON-HATE	0.9638	0.9719	0.9678
VOZ	HATE	0.7347	0.6803	0.7065

Table 6: Per-class results of ViAmpleHate (test set, single run).

6. Result Analysis or Error Analysis

02 Effect of Vietnamese Target Extraction

- English NER detects usable targets in only about 0.09% of Vietnamese training comments, so the original AmpleHate target-aware pathway is almost inactive.
- Vietnamese NER plus the target-cue bank increases target coverage to about 18.8–20.0% on ViHSD and 43.0–45.2% on VOZ-HSD.
- Attack-cue coverage is lower, but it provides an additional signal for identifying hostile predicates.
- The stronger gains on VOZ-HSD align with its much higher target-cue coverage.



03

Qualitative Analysis

- ViAmpleHate helps most when a clear target appears together with an attack predicate.
- Profanity without a group target is often not hate, so separating target and attack cues helps reduce confusion.
- The model still struggles with sarcastic or implicit hate when there is no explicit target or attack cue.
- Neutral mentions of group labels can also be difficult because a target cue alone does not always imply hate.
- These cases show why target-aware modeling is useful but not sufficient for all hate speech patterns.

6. Result Analysis

Comment type	Tgt	Atk	Gold
Group label + hostile predicate	yes	yes	HATE
Profanity, no group target	no	yes	NON
Sarcastic/implicit, no overt cue	no	no	HATE
Group label in neutral context	yes	no	NON

Table 5: Illustrative cases (constructed; replace with real examples). Tgt/Atk = target/attack cue present.



Error Analysis

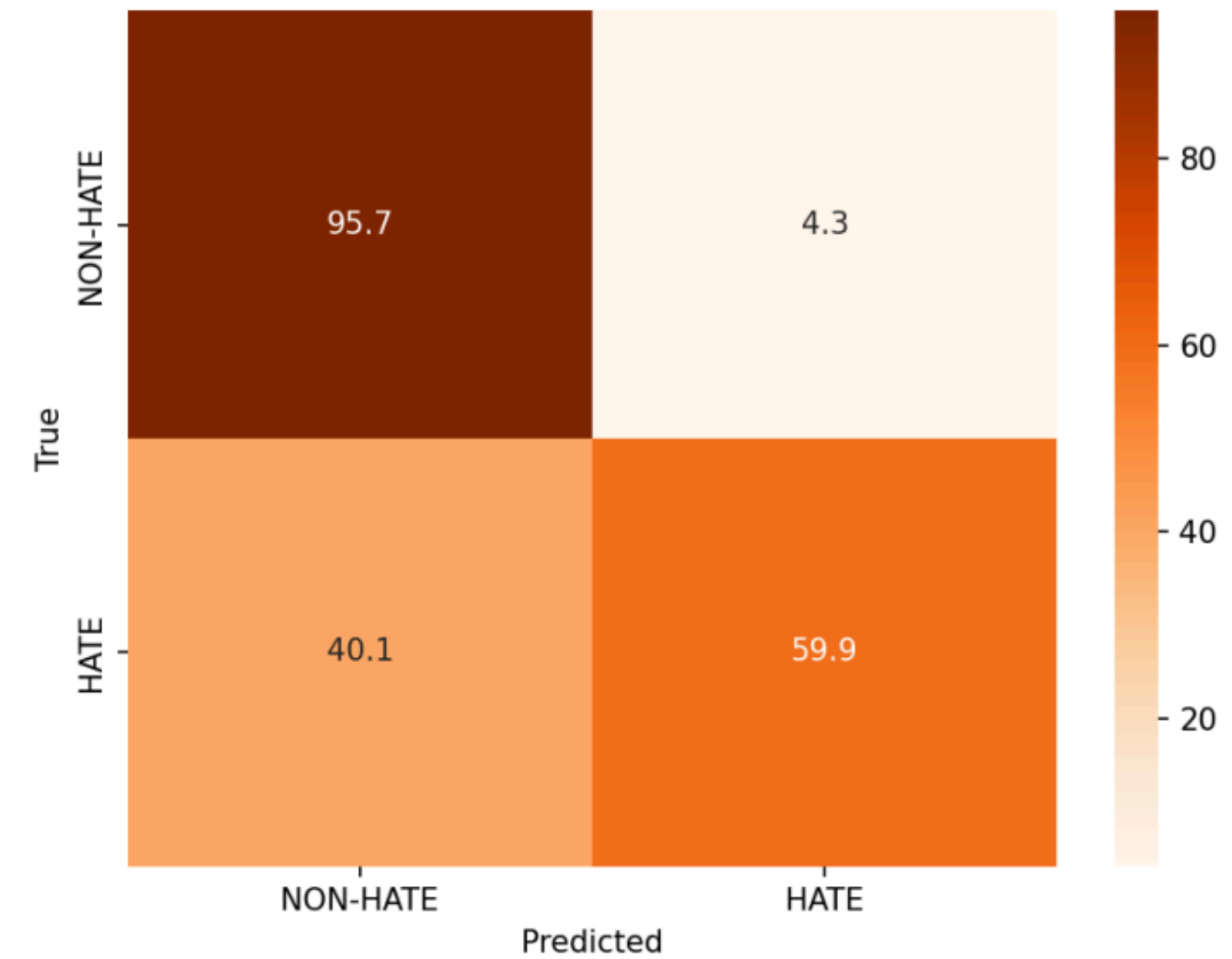
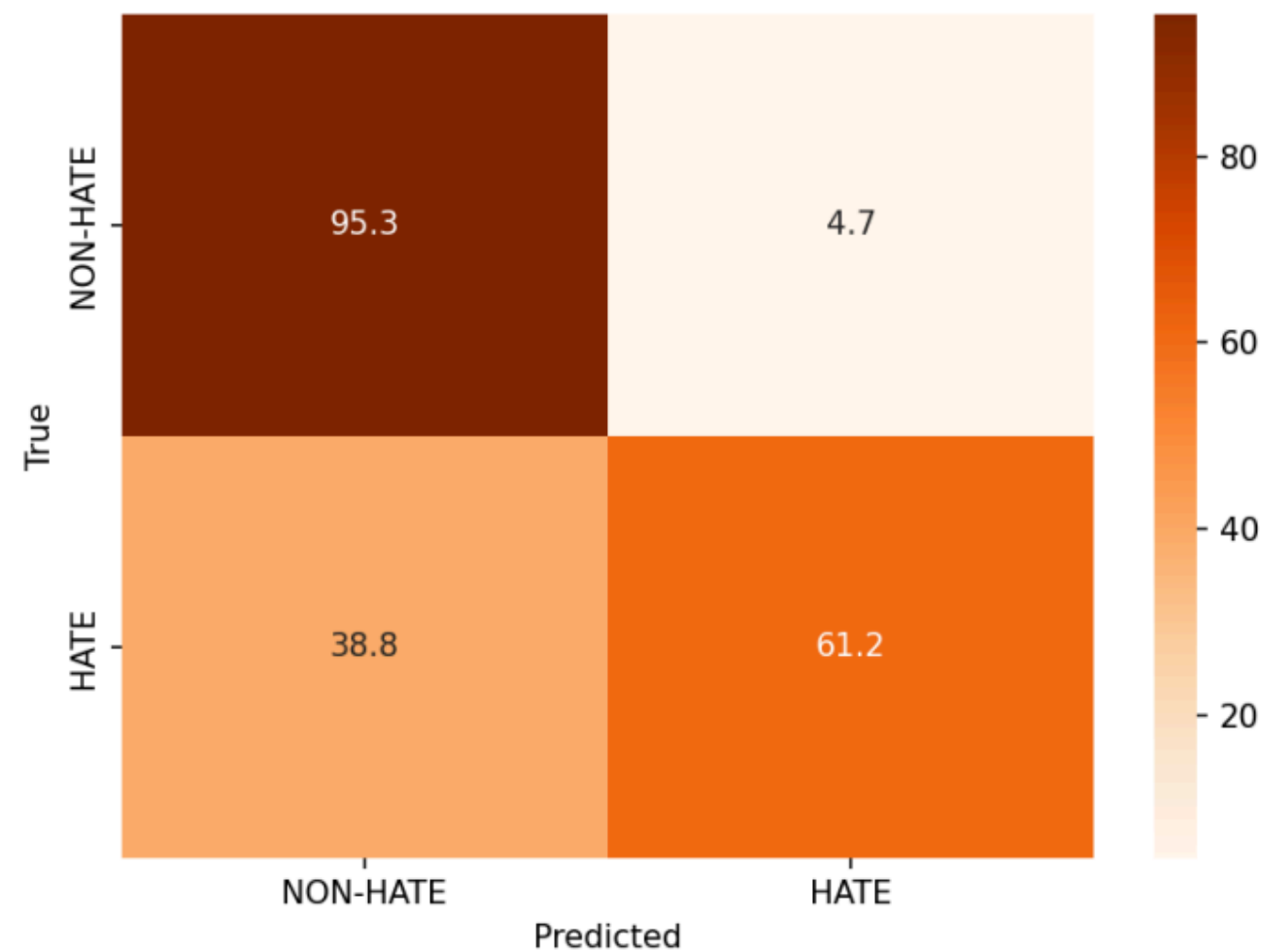


Figure 4: Confusion matrices on ViHSD: baseline AmpleHate (left) vs ViAmpleHate (right). `confusion_matrix_amplehate.png` and `confusion_matrix_viamplehate.png`; note the reduction in HATE false positives.



Error Analysis

The remaining errors fall into seven main categories:



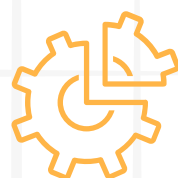
- Offensive but not hate: profanity or insults aimed at individuals can be mistaken as HATE if attack cues are over-weighted.



- Implicit hate: sarcasm, stereotypes, or coded language may express hate without explicit targets or attack words, so cue extraction often fails.



- Ambiguous target references: pronouns, group nouns, and regional labels can appear in neutral or humorous contexts, causing false positives.



- Incomplete cue coverage: fixed cue banks miss new slang, spelling variants, abbreviations, and censored profanity.
- Tokenization/span-alignment errors: NER spans, word segmentation, and PhoBERT subwords may not align correctly, especially for multi-word expressions.
- Threshold sensitivity: a validation-tuned threshold may not transfer well when the test domain changes.
- Class imbalance: HATE examples are only about 10%, so minority-class errors remain difficult despite weighted loss and threshold tuning.

Main improvements suggested: expand cue banks, log per-instance errors with gate value and confidence, add span supervision or sarcasm detection, and apply probability calibration.



Limitations

The remaining errors fall into seven main categories:



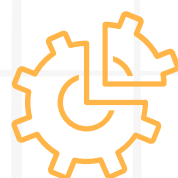
- Cue bank coverage: The target and attack cue banks are partly hand-built, so they cannot fully cover Vietnamese slang, spelling variants, or creative profanity. This limits recall on implicit or novel hate speech.



- Preprocessing dependency: Model performance depends on Vietnamese NER, word segmentation, and tokenization. Errors in these steps may cause cues to be misplaced or missed.



- Threshold limitation: The decision threshold is tuned on the validation set and may not remain optimal under distribution shift. Reporting one threshold also does not show the full precision–recall trade-off.



- Experimental limitation: Results are based on a single run with seed 42, without multi-seed variance, significance testing, or component-level ablation. Therefore, the small ViHSD improvement is not firmly established.

- Comparison limitation: The study compares against re-implemented baselines rather than published ViHSD results.

- Task scope: The model only handles binary NON-HATE/HATE classification and does not yet support multi-target or graded hate detection.



7. Conclusion and Future Work

Conclusion

- ViAmpleHate adapts AmpleHate to Vietnamese hate speech detection by replacing English-oriented target extraction with Vietnamese NER and cue banks.
- It separates target and attack cues, uses relation-bank attention, applies an adaptive relation gate, and trains with weighted cross-entropy plus contrastive loss.
- The model improves Macro-F1 and HATE-F1 over AmpleHate and other baselines, with the clearest gain on VOZ-HSD.
- The ViHSD improvement is small and should be confirmed with controlled multi-seed experiments.
- The coverage analysis supports the idea that target-aware hate speech detection must be adapted to the linguistic surface of each language.

Future Work

- Future work should expand cue banks, mine cues automatically with manual validation, analyze errors using gate values and confidence, add span supervision and sarcasm detection, include discourse context, extend to multi-label and multi-target settings, and improve calibration for cross-domain deployment.



THANK FOR YOUR LISTENING

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